

Configuration Space

Lecture notes – Autonomous & Mobile Robotics, Lecture 1

Contents

1	Configuration Space and Workspace	1
1.1	The idea	1
1.2	Robots and workspaces	2
1.3	Configuration and generalized coordinates	2
1.4	Configuration space	3
2	Examples of Configuration Spaces	3
3	Multi-Body Robots: Manipulators	5
3.1	Planar manipulator with n_b links	5
3.2	Spatial manipulator with n_b links	6
4	Topology of $\mathcal{C}$	6
4.1	The problem with a flat picture of the 2R robot	6
4.2	Wrapping the space: the torus	7
4.3	Manifolds	8
5	Distances in $\mathcal{C}$	8
5.1	What a distance must satisfy	8
5.2	Distances on manifolds and geodesics	9
5.3	The displacement metric	9
5.4	The max over infinitely many points	11
5.5	A practical alternative: control points	12
6	Summary	13
6.1	Configuration spaces seen in this lecture	13
6.2	Distances on $\mathcal{C}$	13
6.3	Key take-aways	13

1 Configuration Space and Workspace

(notes p. 1–2)

1.1 The idea

The central idea is to *define a suitable space in which the robot is a single point*. Once the robot is a point, two core tasks become much simpler to formulate:

- **planning** (find a path for the point from a start to a goal),
- **control** (make the point follow that path).

(Fun fact from the notes: the C-space concept was born to do *collision avoidance*.)

Intuition

Think of a huge, awkwardly shaped sofa that has to be moved through a door. Reasoning about every corner of the sofa at the same time is painful. If instead we find a space where the whole sofa is *one dot*, and the “forbidden” places for the dot are exactly the sofa positions that hit a wall, then moving the sofa becomes moving a dot around obstacles.

1.2 Robots and workspaces

Definition 1.1 (Robot and workspace). A *robot* is a collection of n_b rigid bodies (we assume rigid robots) moving in a *workspace* $\mathcal{W} = \mathbb{R}^2$ or $\mathcal{W} = \mathbb{R}^3$, the physical space in which the robot moves.

- If $\mathcal{W} = \mathbb{R}^2$ the robot is *planar*; if $\mathcal{W} = \mathbb{R}^3$ it is *spatial*.
- If $n_b = 1$ the robot is *single-body*; if $n_b > 1$ it is *multi-body*.

Example 1.2. A car can be modelled as a single body, whereas a car with a trailer is a multi-body system ($n_b = 2$).

1.3 Configuration and generalized coordinates

Definition 1.3 (Posture and configuration). The *posture* of a robot is the position of all its points in $\mathcal{W}$. A *configuration* is a *minimal* set of parameters that defines the posture of the robot in $\mathcal{W}$. The corresponding vector

$$q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$$

is called a *configuration vector*, and each q_i is a *generalized coordinate*.

The generalized coordinates are of two kinds:

- **Cartesian coordinates:** identify the position of representative points of the robot; they take values in $\mathbb{R}^2$ or $\mathbb{R}^3$.
- **Angular coordinates:** identify the orientation of the bodies of the robot; they take values in $\text{SO}(2)$ (planar: one parameter) or $\text{SO}(3)$ (spatial: three parameters).

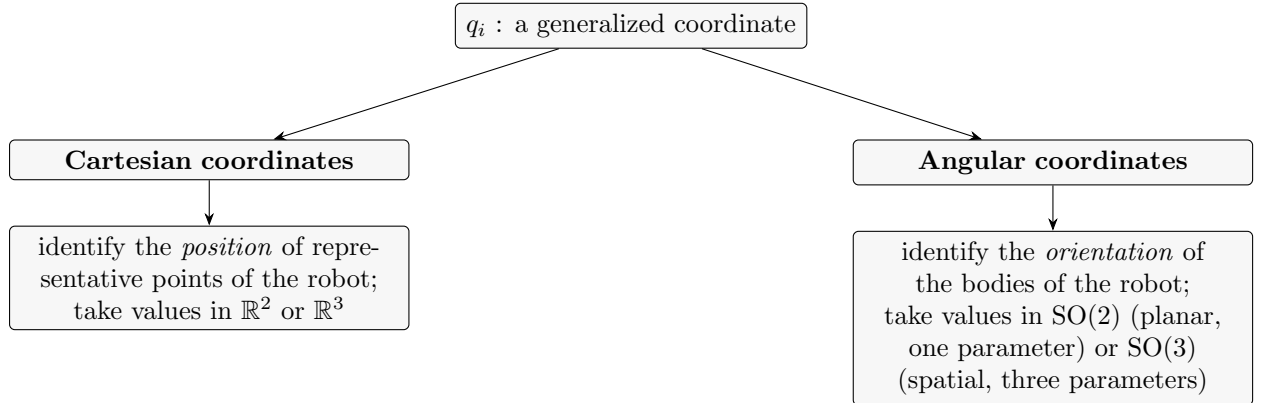


Figure 1: The two kinds of generalized coordinates.

Intuition

“Minimal” is the key word. The posture is the full, redundant description (every point of every body); the configuration is the shortest list of numbers from which the posture can be reconstructed. A rigid rectangle in the plane has infinitely many points, but three numbers (x, y, θ) tell you where all of them are.

1.4 Configuration space

Definition 1.4 (Configuration space). The *configuration space* (C-space) $\mathcal{C}$ of a robot is the set of all possible values of q . Its dimension $n = \dim \mathcal{C}$ is the number of degrees of freedom of the robot.

Intuition

The C-space is the “catalogue of all poses the robot can be in”. Each point of $\mathcal{C}$ is one pose. A curve in $\mathcal{C}$ is therefore a motion of the robot in the physical space, which is exactly why planning for a robot reduces to planning for a point.

2 Examples of Configuration Spaces

(notes p. 2–3)

Example 2.1 (Point robot in $\mathcal{W} = \mathbb{R}^3$). $q = (x, y, z)^\top$, so $\mathcal{C} = \mathbb{R}^3 = \mathcal{W}$ and $n = 3$.

Example 2.2 (Disk robot in $\mathcal{W} = \mathbb{R}^2$). Since a disk looks the same under any rotation, its orientation is irrelevant: $q = (x, y)^\top$, $\mathcal{C} = \mathbb{R}^2 = \mathcal{W}$, $n = 2$.

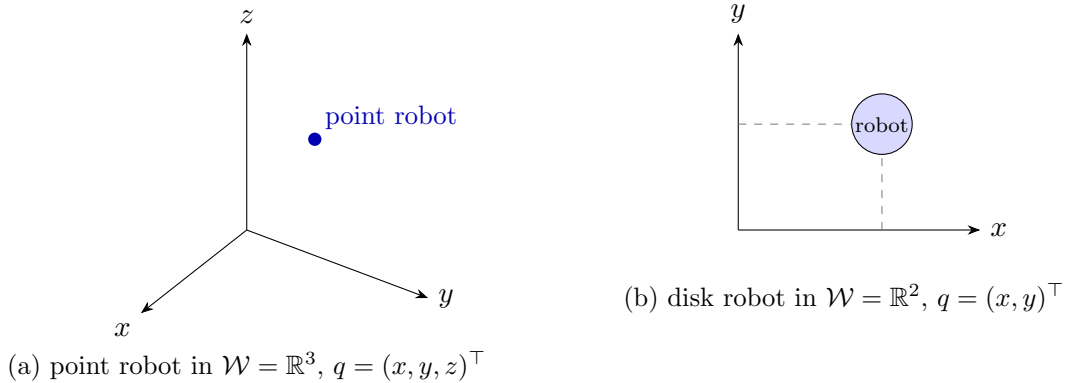


Figure 2: Point robot and disk robot: in both cases $\mathcal{C} = \mathcal{W}$.

In both examples $\mathcal{C}$ and $\mathcal{W}$ coincide. A case where they differ:

Example 2.3 (Disk with a tick). We add a small mark (a “tick”) to the disk, so that its orientation now matters. Then

$$q = \begin{pmatrix} x \\ y \\ \varphi \end{pmatrix}, \quad \mathcal{C} = \mathbb{R}^2 \times \text{SO}(2), \quad n = 3,$$

where $\times$ is the Cartesian product.

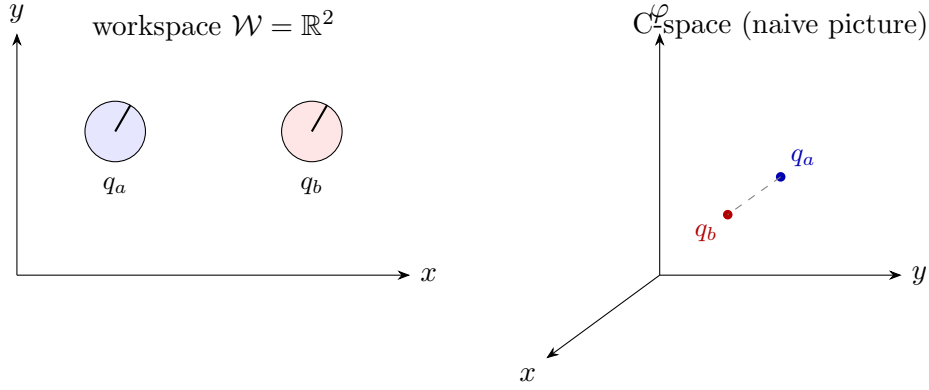


Figure 3: The disk with a tick. Configurations q_a and q_b have the same y and φ but different x . In C-space they differ only along the x -axis. (This picture is *naive*: it is not topologically correct, see Section 4.)

Every change in the configuration space is a path, i.e. a motion, in the physical space.

Example 2.4 (Polygonal robot in $\mathcal{W} = \mathbb{R}^2$). For instance a car moving on the ground. It can rotate about its centre of mass:

$$q = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}, \quad \mathcal{C} = \mathbb{R}^2 \times \text{SO}(2), \quad n = 3.$$

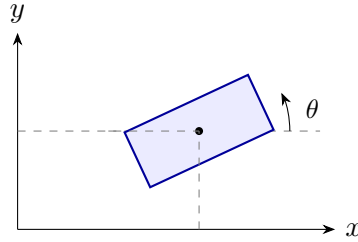


Figure 4: Polygonal robot in the plane: position of the centre of mass (x, y) and orientation θ .

Example 2.5 (Polyhedral robot in $\mathcal{W} = \mathbb{R}^3$). Treat it like the previous case, but add z and two more orientation angles:

$$q = (x, y, z, \alpha, \beta, \gamma)^\top, \quad \mathcal{C} = \mathbb{R}^3 \times \text{SO}(3), \quad n = 6.$$

Definition 2.6 (Special Euclidean groups). The spaces $\mathbb{R}^2 \times \text{SO}(2)$ and $\mathbb{R}^3 \times \text{SO}(3)$ are called the *special Euclidean groups*, denoted $\text{SE}(2)$ and $\text{SE}(3)$.

Intuition

$\text{SE}(n)$ is the set of all rigid motions (a rotation followed by a translation) of n -dimensional space. A free rigid body can be moved anywhere and turned any way, so its configuration space is exactly the set of rigid motions: 3 numbers in the plane (2 translations + 1 rotation), 6 in space (3 translations + 3 rotations).

3 Multi-Body Robots: Manipulators

(notes p. 4-5)

3.1 Planar manipulator with n_b links

Consider a planar manipulator with n_b links and n_b (e.g. revolute) joints moving in $\mathcal{W} = \mathbb{R}^2$. Each link is a rigid body with 3 degrees of freedom in the plane, so one might expect $3n_b$ generalized coordinates. This is wrong: the links are *connected*, and every joint introduces constraints.

Proposition 3.1 (Degrees of freedom of a planar serial arm). *A planar arm of n_b links connected to the ground by n_b revolute joints has*

$$3n_b - 2n_b = n_b$$

degrees of freedom. Each revolute joint imposes 2 constraints (the two coordinates of the joint point must coincide on the two bodies it connects).

Hence, to describe the configuration we do not need to remember all the constraints: one angle per joint is enough,

$$q = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_{n_b} \end{pmatrix}, \quad n = n_b, \quad \mathcal{C} = \underbrace{\text{SO}(2) \times \cdots \times \text{SO}(2)}_{n_b \text{ times}} = \text{SO}(2)^{n_b}.$$

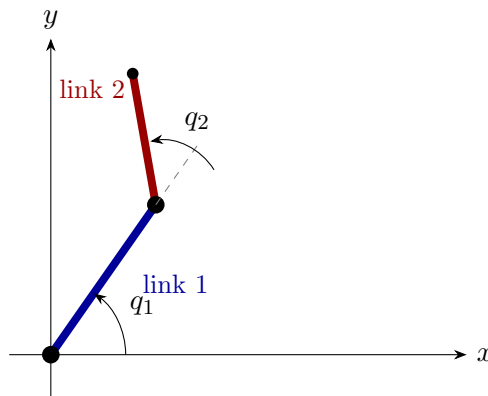


Figure 5: A 2R planar arm: one angle per joint is enough to describe the configuration.

Intuition

Imagine gluing the links together at the joints. Each glued joint removes two “free sliding” directions. Three links loose in the plane would have 9 degrees of freedom; pin them into a chain anchored at the base and only 3 remain: the three joint angles. The number of angles = the number of joints.

Proposition 3.2. *For three revolute joints, $\mathcal{C} = \text{SO}(2) \times \text{SO}(2) \times \text{SO}(2)$, which is very different from $\text{SO}(3)$: the two spaces have the same dimension 3 but different topology.*

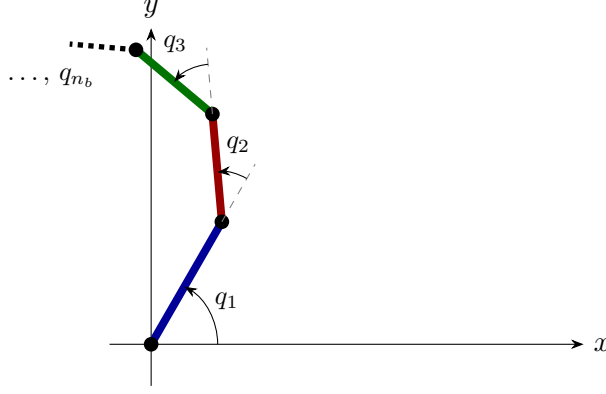


Figure 6: Planar manipulator with n_b links: one joint angle q_i per joint.

Intuition

$\text{SO}(3)$ describes all orientations of *one* rigid body (roll, pitch, yaw). Three planar joints are three *independent* angles each living on its own circle; they cannot all mix into one 3D rotation. Same number of parameters, different shape: it is the shape (topology), not just the count, that matters later.

3.2 Spatial manipulator with n_b links

Proposition 3.3 (Degrees of freedom of a spatial serial arm). *A spatial arm with n_b links and n_b revolute joints in $\mathcal{W} = \mathbb{R}^3$ has*

$$6n_b - 5n_b = n_b$$

degrees of freedom: each link has 6 degrees of freedom, but a revolute joint imposes 5 constraints (3 positions and 2 orientations are fixed; only rotation about the joint axis remains).

Even though the manipulator is spatial, each joint has a *planar* orientation (a single angle), so again

$$\mathcal{C} = \text{SO}(2) \times \cdots \times \text{SO}(2) = \text{SO}(2)^{n_b} \quad (n_b \text{ times}).$$

Intuition

A hinge (revolute joint) always allows exactly one motion: turning about its axis. Whether the world is 2D or 3D, that one motion is a single angle. The ambient dimension changes the *constraint count* (2 vs. 5) but not the end result: one angle per joint.

4 Topology of $\mathcal{C}$

(notes p. 5–6)

4.1 The problem with a flat picture of the 2R robot

Take the 2R planar robot, $\mathcal{C} = \text{SO}(2) \times \text{SO}(2)$, $q = (q_1, q_2)^\top$. Since angles are periodic, q_1 and $q_1 + 2\pi$ give the *same posture*. So if we naively draw $\mathcal{C}$ as the plane $\mathbb{R}^2$ of angle pairs, then two different points give the same posture: the map from this plane to postures is *not injective*, which is bad for us.

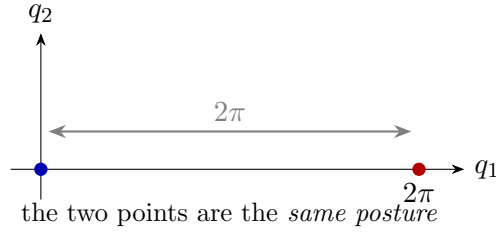


Figure 7: On the plane of angle pairs, $q_1 = 0$ and $q_1 = 2\pi$ (same q_2) correspond to the same posture: the map to postures is many-to-one.

A new attempt: restrict each angle to $[0, 2\pi)$, i.e. draw $\mathcal{C}$ as the square $[0, 2\pi) \times [0, 2\pi)$. Now the correspondence with postures is one-to-one (considering only the two “bad sides” of the square, $q = 0$ and $q = 2\pi$, as the same). But this is still not good enough: two configurations q_a and q_b that are almost the same posture (one just below 2π , the other just above 0) look *far apart* in the square.

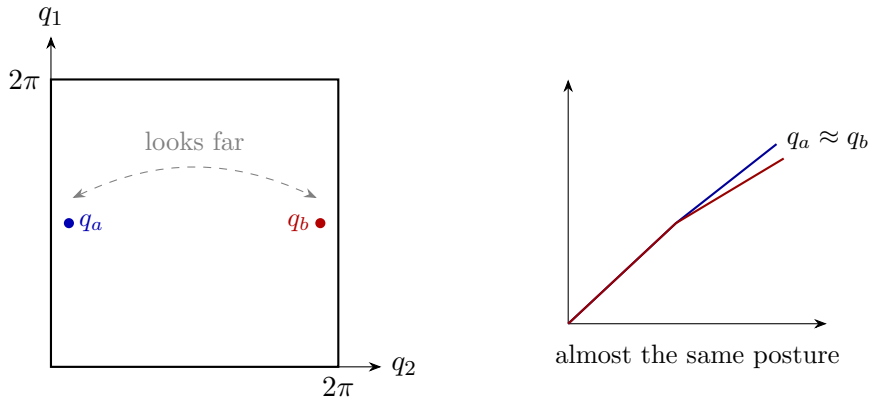


Figure 8: The square picture of $\mathcal{C}$: q_a and q_b are close as postures but far in the flat picture.

So this topology does not describe exactly what is happening.

4.2 Wrapping the space: the torus

The fix is to *wrap* the space: identify the opposite sides of the square (left with right, bottom with top) and embed the resulting 2D object in 3D. We obtain a *torus* (from Latin, “pillow”).

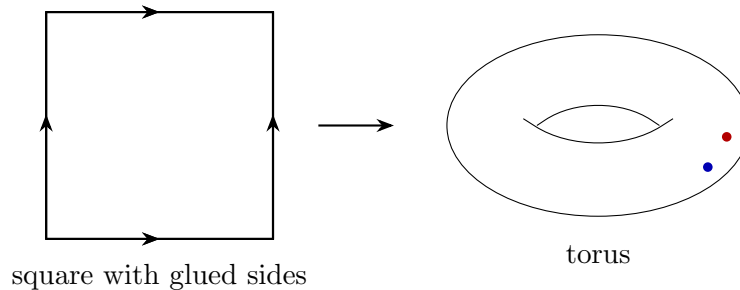


Figure 9: Gluing opposite sides of the square gives a torus. Points that were far apart in the square (across the seam) become neighbours on the torus.

Proposition 4.1. $\text{SO}(2) \cong \mathbb{S}^1$ (the circle), hence the configuration space of the 2R planar robot is $\mathbb{S}^1 \times \mathbb{S}^1 = \mathbb{T}^2$, the torus, and in general $\text{SO}(2)^{n_b} \cong \mathbb{T}^{n_b}$ is an n_b -dimensional torus.

Intuition

A joint angle is like a clock hand: after 2π you are back where you started, so one angle lives on a *circle*, not on a line. Two joints give two circles, and a product of two circles is the surface of a doughnut. Walking off the right side of the square and reappearing on the left is just going around the doughnut the short way.

4.3 Manifolds

Definition 4.2 (Manifold). A *manifold* of dimension n is a space in which the neighbourhood of each point “looks like” a neighbourhood of $\mathbb{R}^n$. (For the torus, $\mathbb{R}^2$.)

Proposition 4.3. Configuration spaces are manifolds (in general not Euclidean anymore): $\mathbb{R}^m$, $\text{SO}(2)$, $\text{SO}(3)$, $\text{SE}(2)$, $\text{SE}(3)$ and their products are all manifolds.

Intuition

The Earth is the classical example: it is a sphere, yet a small enough patch looks flat, so a city map works. Locally we can use ordinary coordinates and calculus; globally the space can wrap around, and that is what a flat picture cannot show.

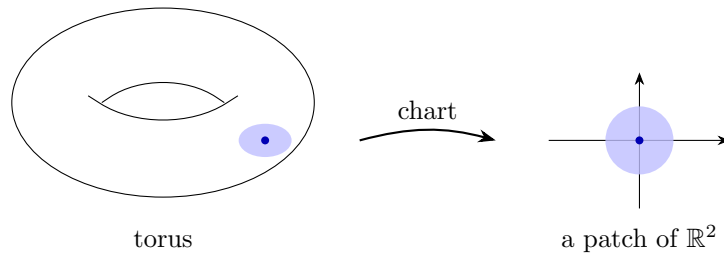


Figure 10: A small neighbourhood of any point of the torus can be flattened (charted) onto a neighbourhood in $\mathbb{R}^2$: this is what “looks like $\mathbb{R}^n$ ” means.

5 Distances in $\mathcal{C}$

(notes p. 6–8)

5.1 What a distance must satisfy

Definition 5.1 (Distance (metric)). A function $d : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}$ is a *distance* if for all $q_A, q_B, q_C \in \mathcal{C}$:

- (i) $d(q_A, q_B) \geq 0$;
- (ii) $d(q_A, q_B) = 0$ if and only if $q_A = q_B$;
- (iii) $d(q_A, q_B) = d(q_B, q_A)$ (symmetry);
- (iv) $d(q_A, q_B) \leq d(q_A, q_C) + d(q_C, q_B)$ (triangle inequality).

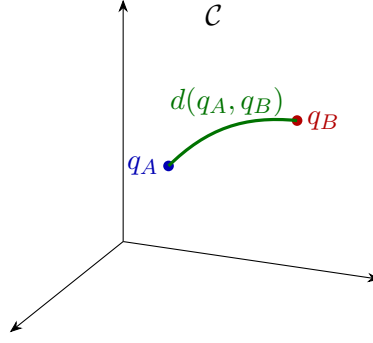


Figure 11: The distance $d(q_A, q_B)$ between two points of the configuration space.

5.2 Distances on manifolds and geodesics

The classic (Euclidean) distance completely ignores the curvature of the space: for two points on a torus, the straight segment between them passes *through the hole*, outside the manifold. So we cannot use the Euclidean distance.

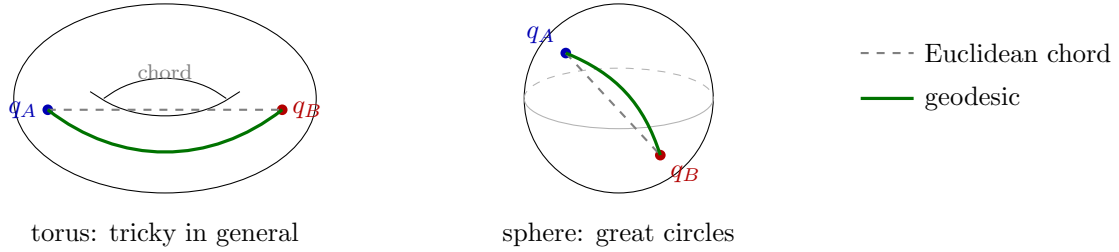


Figure 12: The chord between two points leaves the manifold (it crosses the hole of the torus, or the inside of the sphere); the geodesic stays on the surface. On a sphere the geodesic is an arc of a great circle; on a general manifold it is hard to compute.

Definition 5.2 (Geodesic). A *geodesic* is a minimum-length path between two points *on the manifold*. The geodesic distance is the length of that path.

Intuition

For two cities on Earth, the geodesic is the great-circle route an aircraft follows, not the tunnel through the Earth. Geodesics are the “honest” distances on a curved space.

5.3 The displacement metric

Geodesics are easy on a sphere but can be tricky in general (and on a torus). Because of this difficulty, roboticists do not use them. They use instead a *displacement metric*.

Let p denote a point of the robot (e.g. the centre of mass, or the end point of a link), and $p(q)$ its position in $\mathcal{W}$ when the configuration is q .

Definition 5.3 (Displacement metric).

$$d(q_A, q_B) = \max_{p \in \text{robot}} \|p(q_A) - p(q_B)\|.$$

It is the *maximum displacement* of the points of the robot when moving from q_A to q_B .

Proposition 5.4. *The displacement metric satisfies the four axioms of a distance (provided distinct configurations give distinct postures, which is the case for a minimal parametrization).*

Proof. (i),(iii) hold because a norm is non-negative and symmetric, and the maximum of such quantities is too. (ii) If $q_A = q_B$ every point has zero displacement, so $d = 0$; conversely $d = 0$ means every point is at the same place in the two configurations, i.e. the postures coincide, hence $q_A = q_B$. (iv) For every point p ,

$$\|p(q_A) - p(q_B)\| \leq \|p(q_A) - p(q_C)\| + \|p(q_C) - p(q_B)\| \leq d(q_A, q_C) + d(q_C, q_B);$$

taking the maximum over p on the left gives the claim. $\square$

Intuition

Instead of measuring a length *inside* the curved configuration space, we go back to the physical world and ask: “how far does the worst-behaved point of the robot have to travel?” We use ordinary Euclidean distances there, where they are perfectly fine, and the curvature of $\mathcal{C}$ takes care of itself. Nice side effect: this measure is automatically periodic, since q_1 and $q_1 + 2\pi$ give the same positions.

Example 5.5 (1R robot). A single link of length L rotating about the origin; q_A, q_B are two angles differing by $\Delta \in [0, \pi]$. A point at distance $r \leq L$ from the joint moves by the chord $2r \sin(\Delta/2)$, largest at the tip, so

$$d(q_A, q_B) = 2L \sin\left(\frac{\Delta}{2}\right).$$

We are effectively using the *Euclidean* distance between the two tip positions to measure distance on the manifold.

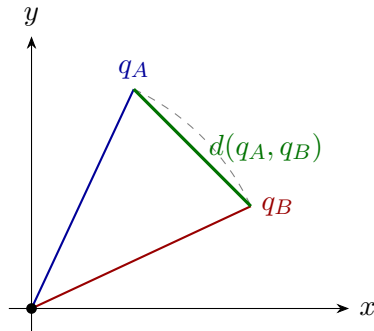


Figure 13: 1R robot: the displacement metric is the Euclidean distance between the tip in the two configurations.

Example 5.6 (Numbers for the 1R robot, $L = 1$). For $\Delta = \pi/2$: $d = 2 \sin(\pi/4) = \sqrt{2} \approx 1.41$. For two angles that differ by $2\pi - 0.1$ (almost a full turn): the wrapped difference is 0.1, so $d = 2 \sin(0.05) \approx 0.10$ – correctly tiny, whereas the flat difference of angles (6.18) would call them far apart.

Example 5.7 (2R robot). With two links we take the maximum over the points of both links. The displacement of a point on a link is an affine function of its position along the

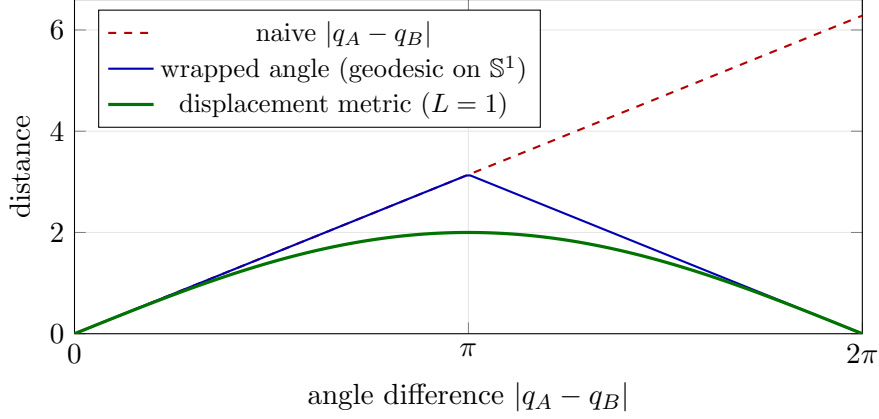


Figure 14: Three ways to measure the distance between two configurations of a 1R robot. The naive difference grows up to 2π ; the other two vanish again at 2π because the configuration wraps around.

link, so its norm is convex and the maximum is attained at a link endpoint: the base (which never moves), the elbow joint, or the tip. In Figure 15 (link lengths 1.6 and 1) the tip moves by about 0.64 but the elbow moves by about 1.60, so $d(q_A, q_B) \approx 1.60$: the maximum is *not* necessarily at the tip.

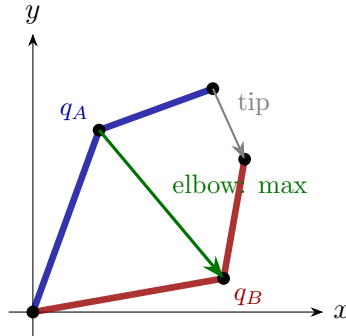


Figure 15: 2R robot in two configurations. The displacement of each characteristic point is drawn; the displacement metric takes the largest one (here the elbow).

5.4 The max over infinitely many points

There is a catch: the definition asks for a maximum over an *infinite* set of points. Intuitively, for polygonal robots we should only look at the *vertices*, but it is not always like this.

Proposition 5.8. *If the robot is a convex polygon (or a union of segments, as a chain of links), the maximum in the displacement metric is attained at a vertex (link endpoint).*

Proof. Under a rigid motion, the position of a body point r is affine in r , so $r \mapsto p(q_A) - p(q_B)$ is affine on the body. The norm of an affine function is convex, and a convex function on a polytope attains its maximum at a vertex. $\square$

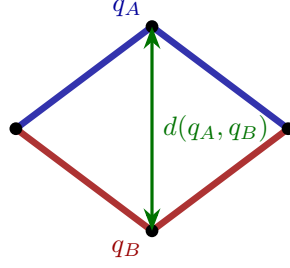


Figure 16: 2R robot with equal links, "elbow up" vs "elbow down", same tip: the tip does not move, and the displacement metric equals the distance between the two elbows.

Intuition

The point that moves most is the one sticking out the farthest – a corner, or the tip of a link. For a curved body (e.g. a disk with attached parts) there is no finite list of corners to check, so we need an approximation.

5.5 A practical alternative: control points

We can use a predetermined number of points, called *control points*, which, if chosen correctly, give a good approximation.

Definition 5.9 (Control-point distance). Fix control points $\{p_1, p_2, \dots, p_c\}$ on the robot. Define

$$d(q_A, q_B) = \sum_{i=1}^c \|p_i(q_A) - p_i(q_B)\|.$$

Algorithm 5.10 (Computing the control-point distance). 1. Choose c representative points $p_1, \dots, p_c$ on the robot (e.g. centre of mass, link endpoints).
2. For each i , compute the forward kinematics $p_i(q_A)$ and $p_i(q_B)$.
3. Return $\sum_{i=1}^c \|p_i(q_A) - p_i(q_B)\|$.

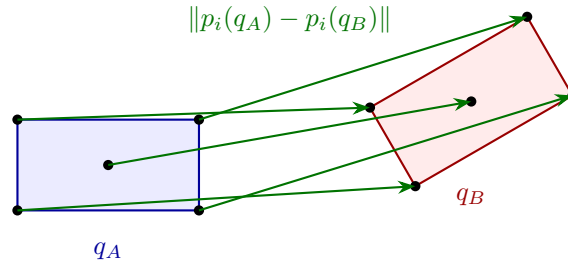


Figure 17: Control points $p_1, \dots, p_5$ (four corners and the centre) of a rectangular robot. The distance is the sum of the lengths of the displacement arrows.

Proposition 5.11. *The control-point distance is symmetric and satisfies the triangle inequality. It is a genuine distance if the control points are chosen so that $p_i(q_A) = p_i(q_B)$ for all i implies $q_A = q_B$; otherwise it is only a pseudo-distance.*

Proof. Each summand is a norm of a difference, so symmetry holds, and the triangle inequality holds termwise and is preserved by the sum. The condition on control points gives axiom (ii). $\square$

Intuition

Rather than tracking every point, you pin a few “sensors” on the robot (say, the tip and the middle) and add up how far they each move. A moving robot cannot fool three well-placed sensors: if they all end up in the same places, the robot is in the same pose. The cost: an approximation of the true worst-case displacement, but cheap and easy to differentiate.

6 Summary

6.1 Configuration spaces seen in this lecture

Robot	$\mathcal{C}$	$n = \dim \mathcal{C}$	Euclidean?
Point in $\mathbb{R}^3$	$\mathbb{R}^3$	3	yes
Disk in $\mathbb{R}^2$	$\mathbb{R}^2$	2	yes
Disk with tick / polygon in $\mathbb{R}^2$	$\text{SE}(2) = \mathbb{R}^2 \times \text{SO}(2)$	3	no
Polyhedron in $\mathbb{R}^3$	$\text{SE}(3) = \mathbb{R}^3 \times \text{SO}(3)$	6	no
Planar n_b -R arm	$\text{SO}(2)^{n_b} = \mathbb{T}^{n_b}$	n_b	no
Spatial n_b -R arm	$\text{SO}(2)^{n_b} = \mathbb{T}^{n_b}$	n_b	no

6.2 Distances on $\mathcal{C}$

Distance	Definition	Pros / cons
Euclidean in parameters	$\ q_A - q_B\ $	simple, but ignores wrapping (bad on angles)
Geodesic	shortest path on the manifold	exact, but hard to compute in general
Displacement metric	$\max_p \ p(q_A) - p(q_B)\ $	respects topology, physically meaningful; max over infinitely many points
Control points	$\sum_i \ p_i(q_A) - p_i(q_B)\ $	cheap approximation; needs well-chosen points

6.3 Key take-aways

- The configuration space $\mathcal{C}$ is the space of all values of a *minimal* set of parameters describing the robot; in it the robot is a single point, and planning/control become problems about that point.
- Joints reduce degrees of freedom: a planar arm has $3n_b - 2n_b = n_b$ and a spatial arm $6n_b - 5n_b = n_b$ degrees of freedom – one angle per joint.
- Because angles are periodic, $\mathcal{C}$ is generally a *manifold* (e.g. a torus for a 2R arm), not $\mathbb{R}^n$: same dimension does not mean same shape ($\text{SO}(2)^3 \neq \text{SO}(3)$).
- Euclidean distance in parameter space is misleading on a manifold; geodesics are correct but hard to compute.
- Roboticians use the *displacement metric* (max point displacement) or its practical version with a few *control points*.